# GENESIS OS / RELATIONSHIP SCORES

Sanctuary Core Module State Machine & Payload Specification v1

Developer contract for Sanctuary runtime objects, submode flows, logging packets, AI permissions, write-back law, and ExsAnonymous integration

- Purpose. Convert Sanctuary from feature vision into strict engineering packets so modules behave as governed software objects rather than loose emotional surfaces.
- Authority. Built to sit beside the corrected master blueprint and workbook-aligned system. Sanctuary never overrides workbook score law; it enriches evidence, interventions, memory, and forecast context under governance.
- Important correction. ExsAnonymous is treated in this specification as a first-class Sanctuary module family, not a detached companion feature.

References used: Corrected Master Blueprint sections on Sanctuary, AI boundaries, telemetry, companion evidence, and recovery modules; workbook governance model and source-of-truth rules.

## 1. Purpose, Scope, and Non-Negotiable System Stance

This specification defines Sanctuary as a governed runtime family composed of explicit modules, submodes, state machines, payload contracts, logging packets, write-back rules, safety controls, AI permissions, and recovery paths. It is written so product, backend, mobile, web, data, trust and safety, and AI orchestration teams can build the same behavior without improvisation.

- Sanctuary is a protected reflective layer for users, not a free-form score mutator.
- Assessment and workbook logic remain anchor truth for stable identity variables and public score law.
- Sanctuary may generate evidence, confidence changes, intervention prompts, archive entries, drift confirmations, cluster seeds, and forecast context under capped influence.
- No single Sanctuary session may rewrite archetype, shadow, readiness, or pair forecast directly without passing the controlled write-back path defined here and in readiness and forecast contracts.
- ExsAnonymous is inside Sanctuary governance and uses the same session object, payload versioning, safety model, and archive law as all other Sanctuary modules.

## 2. Source-of-Truth Alignment

- Corrected Master Blueprint owns product scope, architecture, AI boundaries, telemetry posture, and the retained Sanctuary logging object.
- Workbook owns score law, compatibility runtime order, controlled labels, and the rule that companion evidence enriches confidence rather than replacing deterministic pair computation.
- Readiness and Forecast contracts own what Sanctuary may influence, when it may influence it, and how output suppression works.
- Cluster Rule Pack Specification and cluster files own how Sanctuary evidence seeds, confirms, escalates, or clears clusters.
- This Sanctuary specification owns runtime module behavior, state transitions, payloads, logging packets, AI permissions, write-back classes, and safety enforcement.

## 3. Sanctuary Module Registry

| Module ID | Module Name | Primary Scope | AI Mode | Primary Outputs |
|---|---|---|---|---|
| SAN_STILL | Stillness Mode | single-user | optional assist | regulation event, emotional delta, intervention history |
| SAN_MIRROR | Emotional Mirror | single-user | guided | reflection packet, cluster seed, |

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|---|---|---|---|---|
| | | | | archive note |
| SAN_WHISPER | Whisper Window | single-user / match-context | assisted compose | draft object, unsent reflection, pattern signal |
| SAN_ARCHIVE | Archive of Self | single-user memory | summarize only | journal record, retrieval index, trend line |
| SAN_COMPASS | Alignment Compass | single-user trend | optional summarize | state quadrant, drift context, prompt routing |
| SAN_RITUAL | Micro Rituals | single-user | guided or static | ritual completion, reset outcome, repeat cadence |
| SAN_GHOST | Ghost Mode | single-user privacy state | no proactive AI | visibility change, pool suppression, reflection gate |
| SAN_LASTLIGHT | Last Light | grief-linked single-user | guided with safeguards | healing session, grief overlay, archive echo |
| SAN_EXS | ExsAnonymous | single-user healing / pattern extraction | guided with safeguards | ex narrative packet, residue signals, cluster seeds |
| SAN_DMREF | Sanctuary-DM Reflection | single-user with match context | guided | conversation reflection, relational pattern packet |
| SAN_TRUTH | Truth Signal | single-user daily insight | generated | one-line insight, module routing |
| SAN_MM_GATE | Mirror Matches Reflection Gate | single-user pre-match gate | guided minimal | gate completion, intention check, visibility unlock |

## 4. Universal Sanctuary State Machine

Every Sanctuary module must implement the shared top-level state vocabulary below. Modules may add local sub-states, but they may not remove or rename the universal states because logs, AI orchestration, and QA rely on normalized transitions.

| State | Meaning | Entry Conditions | Legal Outbound Transitions |
|---|---|---|---|
| IDLE | Module not active and no current session exists. | User eligible but has not entered module. | AVAILABLE, SUPPRESSED |
| AVAILABLE | Module can be entered. | Policy, consent, and gating pass. | INITIATED, SUPPRESSED |
| INITIATED | Session shell created; pre-checks running. | User taps entry or system routes into module. | AWAITING_INPUT, ACTIVE, FALLBACK_STATIC, ERRORED |
| AWAITING_INPUT | Module expects user action. | Prompt shown; no sufficient response yet. | ACTIVE, PAUSED, CANCELLED, ERRORED |
| ACTIVE | Primary module experience in progress. | Required payload present and checks passed. | AWAITING_AI, AWAITING_CONFIRMATION, COMPLETED, PAUSED, SAFETY_HOLD, ERRORED |
| AWAITING_AI | AI job requested and pending. | AI permission enabled and content queued. | ACTIVE, AWAITING_CONFIRMATION, FALLBACK_STATIC, SAFETY_HOLD, ERRORED |
| AWAITING_CONFIRMATION | User must confirm save, send, or close step. | Draft or interpretation created. | COMPLETED, ACTIVE, CANCELLED |
| PAUSED | Session intentionally paused or interrupted. | App backgrounded, timeout, or manual pause. | ACTIVE, CANCELLED, EXPIRED |
| COMPLETED | Session closed successfully. | Required write-backs and logs committed. | ARCHIVED, AVAILABLE |
| ARCHIVED | Session persisted to archive or analytics store. | Completion packet finalized. | AVAILABLE |
| SUPPRESSED | Entry blocked or | Safety, privacy, | AVAILABLE |

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|---|---|---|---|
| | downgraded by policy. | eligibility, or insufficient context failed. | |

## 5. Universal Session Object and Logging Packet

The corrected blueprint already defines a Sanctuary logging object. This specification hardens it into a canonical packet. Every module session must emit one root session packet and one event packet per meaningful transition.

| Packet | Required fields |
|---|---|
| Root session packet | session_id, user_id, module_id, submode_id, session_version, entry_surface, started_at, ended_at, current_state, pre_state_emotion_code, post_state_emotion_code, duration_ms, completion_state, intervention_type, optional pair_id, optional match_context_id, consent_scope, ai_mode, safety_level, payload_version, archive_write_class, created_by |
| Per-event packet | event_id, session_id, event_seq, event_type, source_state, destination_state, timestamp, actor_type (user, system, ai, moderation), prompt_id, payload_ref, tokenized_text_ref, derived_signal_refs, cluster_refs, drift_refs, grief_refs, visibility_scope, failure_code, retry_count, provenance |

- All packets are append-only. Mutations create new versioned packets rather than silently replacing the prior state.
- Text content is stored by reference where possible; logs should point to content objects rather than duplicate long text bodies.
- Module summaries, AI inferences, and pattern reads must include evidence band, freshness, provenance, and consent scope.
- If a module operates in static mode with no AI, the log must still be complete.

## 6. Global Write-Back Classes

| Class | Name | Destination | Rules |
|---|---|---|---|
| WB0 | No durable write-back | none | Transient interaction only; analytics may |

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|---|---|---|---|
| | | | count the event but no durable user-facing record. |
| WB1 | Archive note | Archive of Self | Reflection or ritual record stored; may power retrieval and user history but does not affect readiness or forecast directly. |
| WB2 | Evidence packet | Evidence ledger | Durable evidence for drift, grief, clusters, or confidence; subject to influence caps and thresholds. |
| WB3 | Guidance context | AI context store | Soft context for future coaching; never allowed to mutate score law directly. |
| WB4 | State update | Sanctuary state store | Updates Ghost Mode, Alignment Compass, module preferences, or cadence flags. |
| WB5 | Governed downstream packet | Readiness, forecast, or moderation queue | Only after contract thresholds; used for suppression, gating, or review paths. |

## 7. SAN_STILL - Stillness Mode

Stillness Mode is Sanctuary's lowest-volatility regulation surface. It exists to slow physiological escalation, help the user name or regulate state, and produce a clean before and after emotional delta. It is not a diary, not a score rewrite surface, and not a therapy simulator.

### Entry conditions

- User is signed in and not in a safety-blocked state.
- Module may be entered manually from Sanctuary home, nudged after heavy emotional activity, or suggested after message, cluster, or drift events.
- AI assistance is optional; static exercises must always exist if AI is unavailable or suppressed.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| STILL_INIT | Session shell created | Read pre-state emotion code and selected practice path | BREATH_ACTIVE, AMBIENT_ACTIVE, CHECKIN_ONLY, CANCELLED |
| BREATH_ACTIVE | Timed breathing ritual selected | Run pacing and capture completion markers | REFLECTION_PROMPT, COMPLETED, PAUSED |
| AMBIENT_ACTIVE | Ambient or stillness media selected | Track dwell and optional mood tap | REFLECTION_PROMPT, COMPLETED, PAUSED |
| CHECKIN_ONLY | User chooses minimal check-in | Collect quick emotional self-report | COMPLETED, AWAITING_AI |
| REFLECTION_PROMPT | Post-regulation prompt shown | Collect one-line or scale response | AWAITING_AI, COMPLETED, CANCELLED |
| STILL_ARCHIVE | Optional save to archive | Persist note and delta | COMPLETED |

### Input payload contract

- pre_state_emotion_code (required)
- selected_practice_type: breath | ambient | silent_timer | quick_reset
- optional timer_length_sec
- optional mood_word_before
- optional free_text_before
- optional match_context_id if launched from relationship surface

### Primary outputs

- post_state_emotion_code and emotional delta
- session completion metrics
- optional archive note
- optional evidence packet for acute escalation de-intensified
- cadence history for which regulation tools work best

### Write-back law

- Default is WB1 archive note plus WB4 cadence or state update.
- Only repeated acute dysregulation patterns may create WB2 evidence packets.
- Stillness completion alone never lowers readiness and never changes forecast directly.
- Positive completion history may increase intervention confidence for future routing, not score law.

### AI permissions and constraints

- AI may explain an exercise, summarize the user's own words, and offer a short grounded reflection.
- AI may not diagnose, catastrophize, or infer deep identity variables from one regulation session.
- AI summaries must be optional and suppressible; static non-AI completion path must remain available.

### Safety, suppression, and fallback

- If the user abandons mid-session, log partial completion but do not shame or aggressively reprompt.
- If safety signals spike, transition to SAFETY_HOLD and present safer support language, not deep reflection.
- If AI fails, fall back to static stillness completion without data loss.

### Implementation notes

- Recommended pre and post emotion coding should map to Emotional Hex or approved simplified Sanctuary state labels.
- Timed media assets must be versioned so analytics can compare which formats produce cleaner de-escalation.

## 8. SAN_MIRROR - Emotional Mirror

Emotional Mirror is the interpretive reflection layer. It helps the user articulate what they are feeling, what may be driving it, and what pattern it might belong to. It is allowed to seed clusters, but only under evidence rules.

### Entry conditions

- User enters voluntarily or is routed by check-in, messaging friction, or post-match reflection prompt.
- Consent for reflective AI must be enabled for AI-generated mirrors; a static question path must exist otherwise.
- Mirror may run with or without direct match context.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| MIRROR_INIT | Session created | Load prompt family based on entry surface | PROMPT_SELECT, QUICK_MIRROR, CANCELLED |
| PROMPT_SELECT | Prompt family shown | User selects mood or topic | AWAITING_INPUT |
| AWAITING_INPUT | Mirror asks structured prompt | Collect text, sliders, or tags | MIRROR_PARSE, CANCELLED |
| MIRROR_PARSE | System parses user input | Generate draft reflection packet | AWAITING_AI, REVIEW_DRAFT, SAFETY_HOLD |
| REVIEW_DRAFT | Static or AI draft ready | User reviews and edits | CONFIRM_SAVE, ACTIVE |
| CONFIRM_SAVE | User chooses save depth | Commit archive and evidence routing | COMPLETED |
| MIRROR_ARCHIVE | Archive path selected | Persist reflection artifacts | COMPLETED |

### Input payload contract

- prompt_family_id
- structured emotional tags
- optional narrative_text
- optional target_person_role (self, match, ex, parent, other)
- optional related cluster_refs or drift_refs if launched from existing signal
- optional privacy_level: private | future-ai-context | archive-visible

### Primary outputs

- mirror_reflection_packet
- soft pattern labels or suggested next step
- optional cluster seed with low initial confidence
- optional archive entry
- optional routing recommendation to Stillness, Whisper, ExsAnonymous, or DM Reflection

### Write-back law

- Simple use creates WB1 archive note.
- If Mirror identifies a governed cluster pattern with sufficient evidence, it may write WB2 evidence packet referencing the relevant rule pack.
- Mirror may update Alignment Compass through WB4 if the user confirms quadrant selection or the module uses an approved deterministic mapping.
- Mirror may never directly recalculate readiness or pair forecast.

### AI permissions and constraints

- AI may summarize, reflect, propose hypotheses, and suggest next modules.
- AI must label uncertain language as a possibility, not a verdict.
- AI cannot expose internal risk labels or hidden shadow severity fields.
- Any AI-generated pattern language must be soft-translated for the user and include provenance.

### Safety, suppression, and fallback

- High-risk disclosures, self-harm cues, or severe abuse cues require SAFETY_HOLD and appropriate trust flow.
- If content is too sparse, the module should downshift to simple labeling rather than over-interpreting.
- Do not turn one emotional entry into a permanent personality conclusion.

### Implementation notes

- Emotional Mirror is one of the main front doors for cluster seeding, so its prompt families should be versioned and QA-tested.
- Suggested prompt family examples: What stung, What are you needing, What keeps replaying, What are you protecting.

## 9. SAN_WHISPER - Whisper Window

Whisper Window is the unsent emotional draft and message processing tool. It lets the user say what they want to say, refine it, keep it private, or transform it into a safer communication option. It is not a hidden DM channel and not a license for the AI to speak for the user without consent.

### Entry conditions

- Available from Sanctuary home, chat surfaces, match cards, or post-trigger prompts.
- Requires target context only when the user wants match-specific drafting; private unsent mode requires no target.
- AI drafting requires composition consent.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| WHISPER_INIT | Session created | User selects private draft or target context | DRAFTING, TEMPLATE_PICK, CANCELLED |
| DRAFTING | User writes freely | Capture raw text and tone target if chosen | ANALYZE_DRAFT, SAVE_PRIVATE, CANCELLED |
| TEMPLATE_PICK | User wants prompt or template | Load template family | DRAFTING |
| ANALYZE_DRAFT | System inspects text | Surface tone options and risk checks | AWAITING_AI, REVIEW_OPTIONS, SAFETY_HOLD |
| REVIEW_OPTIONS | Options ready | User compares keep-private, refine, soften, or send-to-chat paths | CONFIRM_ACTION, DRAFTING |
| CONFIRM_ACTION | User chooses final action | Persist or hand off | COMPLETED |
| SAVE_PRIVATE | User keeps it unsent | Archive private whisper | COMPLETED |

### Input payload contract

- raw_draft_text
- optional target_user_id, pair_id, or match_context_id
- optional tone_goal: honest | softer | boundaried | concise | not_send
- optional emotional intensity self-rating
- optional include_context_snippet from chat, if permitted

### Primary outputs

- private whisper object
- edited or softened draft options
- send-safe message suggestion
- pattern hints about escalation, avoidance, overexplaining, or unreadiness
- optional archive note and cluster seeds

### Write-back law

- Private unsent use usually writes WB1 only.
- Repeated drafting-withholding patterns may produce WB2 evidence for relevant clusters only after thresholds.
- If the user chooses to send a message, the actual send belongs to messaging service; Sanctuary stores only the handoff packet and result.
- Whisper output may update AI guidance context through WB3 when consented.

### AI permissions and constraints

- AI may rephrase, shorten, soften, or clarify the user's own message.
- AI may not impersonate the user, contact another person autonomously, or turn text into manipulation advice.
- AI must preserve an option to keep the message private and unsent.

### Safety, suppression, and fallback

- Manipulative, threatening, or coercive content should be refused or redirected with safe language.
- If target context permissions fail, downshift to private draft mode.
- Chat snippets pulled into Whisper require explicit permission and minimal necessary scope.

### Implementation notes

- Useful derived fields: draft_intensity_band, clarity_score, send_intent, withhold_intent, regret_risk estimate.
- Do not expose internal manipulation or shadow flags directly to the user.

## 10. SAN_ARCHIVE - Archive of Self

Archive of Self is the durable memory surface for Sanctuary outputs. It is a governed repository for the user's reflections, rituals, whispers, grief echoes, ExsAnonymous sessions, and trend history. It is not a free-form engine memory dump.

### Entry conditions

- Always available to the owner unless account or privacy state suppresses some records.
- Records enter Archive only through approved module write-back classes and retention rules.
- Retrieval of archived records for AI requires explicit consent scope.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|

| | | | |
|---|---|---|---|
| ARCHIVE_IDLE | Viewer not active | No retrieval in progress | ARCHIVE_VIEW, SEARCH, SUPPRESSED |
| ARCHIVE_VIEW | User browsing entries | Load summaries and filters | ENTRY_DETAIL, SEARCH, EXIT |
| SEARCH | User queries archive | Resolve tags, dates, module families | ENTRY_DETAIL, ARCHIVE_VIEW |
| ENTRY_DETAIL | Specific entry opened | Display source, summary, related records | ARCHIVE_VIEW, EXPORT_REQUEST |
| EXPORT_REQUEST | User requests export | Package permitted records | COMPLETED, ERRORED |

### Input payload contract

- archive filters, tags, date ranges, module family filters
- optional query text
- optional consented AI retrieval request

### Primary outputs

- retrieved archive cards
- trend summaries
- linked module history
- export bundle when allowed

### Write-back law

- Archive is primarily WB1 destination but may store pointers to WB2, WB3, and WB4 packets.
- Archive cannot itself create readiness or forecast changes; it is a retrieval and memory surface.
- AI-generated archive summaries must never replace raw user-authored content.

### AI permissions and constraints

- AI may summarize, cluster, or surface trends across archive entries only with consent.
- AI may not rewrite original entry text or present a generated summary as the only source of truth.
- AI retrieval should be bounded by minimal-needed scope and freshness.

### Safety, suppression, and fallback

- Sensitive content can be hidden, redacted, or excluded from retrieval surfaces per user controls and safety policy.

- Deleted or expired content must respect retention policy and not linger in AI context.
- Archive retrieval failures should not corrupt source entries.

### Implementation notes

- Archive requires strong versioning because many later systems depend on stable references to historical Sanctuary outputs.

## 11. SAN_COMPASS - Alignment Compass

Alignment Compass is the quadrant and trend layer that helps the user place their current inner state in a simple map such as Connected, Drained, Hopeful, or Guarded. It turns repeated reflective data into a legible state history without pretending to be diagnosis.

### Entry conditions

- Available after at least one qualifying Sanctuary check-in or reflective module.
- Can be entered directly or shown as a post-session state selection step.
- May compute suggested quadrant from approved deterministic inputs, but user confirmation should remain primary where possible.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| COMPASS_INIT | Session starts | Load current quadrant and history | QUADRANT_SELECT, TREND_VIEW |
| QUADRANT_SELECT | User selects or confirms quadrant | Capture quadrant and optional note | TREND_VIEW, CONFIRM_SAVE |
| TREND_VIEW | History and signals shown | Render trends and related module hints | CONFIRM_SAVE, EXIT |
| CONFIRM_SAVE | Save chosen | Commit state update and optional archive note | COMPLETED |

### Input payload contract

- quadrant_selection
- confidence_source: self-report | deterministic-suggested | hybrid
- optional note
- optional linked session_id

### Primary outputs

- current_quadrant state

- trend history record
- prompt routing recommendations
- possible drift context enrichment

### Write-back law

- Compass writes WB4 state updates by default.
- Repeated persistent guarded or drained patterns may emit WB2 evidence for drift or cluster support only after threshold rules.
- Compass may contribute to forecast confidence but not directly rewrite pair score.

### AI permissions and constraints

- AI may summarize the trend line and suggest next Sanctuary modules.
- AI may not use poetic certainty to tell the user who they are forever.
- Suggestions must reference whether state is self-reported or machine-suggested.

### Safety, suppression, and fallback

- If insufficient history exists, show sparse state view instead of fake trends.
- If the user declines quadrant labeling, the module should permit skip without coercion.

### Implementation notes

- Quadrant taxonomy must remain controlled and versioned for analytics compatibility.

## 12. SAN_RITUAL - Micro Rituals

Micro Rituals are short emotional reset tools such as breath pacing, grounding prompts, symbolic releases, mini visualizations, and brief written acts. They are sub-5-minute rituals with explicit payloads and completion events.

### Entry conditions

- Available from Sanctuary home, post-trigger suggestions, and after difficult reflective sessions.
- Each ritual type must exist as a versioned template with defined required steps.
- AI may guide, but a static scripted ritual must also exist.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| RITUAL_PICK | Ritual list shown | Select template and duration | RITUAL_ACTIVE, CANCELLED |
| RITUAL_ACTIVE | Ritual running | Deliver steps and track markers | RITUAL_CHECK, PAUSED, CANCELLED |

| | | | |
|---|---|---|---|
| RITUAL_CHECK | Completion or closeout prompt | Collect outcome and repeat intent | COMPLETED, AWAITING_AI |

### Input payload contract

- ritual_template_id
- optional trigger_reason
- optional pre-state emotion code
- optional intent tag: calm | closure | grounding | courage | release

### Primary outputs

- completion packet
- emotional delta if captured
- repeat preference
- archiveable ritual note

### Write-back law

- Usually WB1 plus WB4 cadence preferences.
- Rituals may contribute to intervention effectiveness analytics.
- No ritual alone may alter readiness or forecast.

### AI permissions and constraints

- AI may narrate or personalize a ritual lightly.
- AI may not invent unsafe ritual instructions or present spiritual certainty as fact.

### Safety, suppression, and fallback

- Templates with physical instructions must be safety-reviewed.
- If the user exits early, log incomplete but preserve dignity; no guilt loops.

## 13. SAN_GHOST - Ghost Mode

Ghost Mode is the privacy and emotional shelter control inside Sanctuary. It temporarily removes the user from the visible match pool or reduces interaction surfaces while preserving reflection access. It is a policy state, not a mood itself.

### Entry conditions

- User may toggle voluntarily, or the system may recommend but not auto-force except under explicit safety policy.
- Requires clear explanation of what visibility changes.
- Some pair or couple contexts may require additional confirmation.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| GHOST_VIEW | State pane opened | Show current visibility and effects | TOGGLE_REQUEST, EXIT |
| TOGGLE_REQUEST | User requests enter or exit | Run eligibility and dependency checks | CONFIRM_TOGGLE, SUPPRESSED |
| CONFIRM_TOGGLE | User confirms | Commit visibility change | COMPLETED |
| COOLDOWN_VIEW | User sees cooldown details | Explain timers and return path | EXIT |

### Input payload contract

- requested_state: on | off
- duration_option if supported
- reason_tag optional
- acknowledgement of side effects

### Primary outputs

- visibility state update
- cooldown timers
- match pool suppression or reentry packet
- optional private archive note

### Write-back law

- Ghost Mode primarily writes WB4 state updates and downstream visibility packets.
- Readiness and forecast are not directly reduced by using Ghost Mode.
- Repeated emergency-like toggling may contribute to drift evidence only under explicit thresholds.

### AI permissions and constraints

- AI may explain Ghost Mode in plain language but may not pressure the user to enable or disable it.
- AI should offer reflection support after activation if the user opts in.

### Safety, suppression, and fallback

- Must clearly separate privacy choice from punitive restriction.

- If active conversations exist, the system should explain what remains possible and what is muted.

## 14. SAN_LASTLIGHT - Last Light

Last Light is the grief-linked remembrance and release experience within Sanctuary. It is designed for connection with memory, not supernatural certainty. It must be governed with extra sensitivity and explicit language boundaries.

### Entry conditions

- User voluntarily enters the grief pathway.
- Requires high-sensitivity content policy and reviewed copy library.
- Optional photo, symbol, song, or memorial inputs are permitted under safety and storage rules.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| LL_INIT | Session starts | Select memorial object or create new | MEMORY_INPUT, CANCELLED |
| MEMORY_INPUT | User provides name, symbols, or message target | Capture memorial context | GUIDED_REFLECTION, STATIC_RITUAL |
| GUIDED_REFLECTION | Reflection flow active | Prompt memory, gratitude, release, or unfinished words | AWAITING_AI, RELEASE_STEP, SAFETY_HOLD |
| STATIC_RITUAL | Non-AI ritual flow | Run gentle remembrance sequence | RELEASE_STEP, COMPLETED |
| RELEASE_STEP | Final release or closure step | Collect closing action | ARCHIVE_ECHO, COMPLETED |
| ARCHIVE_ECHO | Save under Echoes | Persist memorial session | COMPLETED |

### Input payload contract

- memorial_name_or_label
- optional media_refs
- selected symbols or songs
- optional message text
- grief intensity self-rating

- privacy setting

### Primary outputs

- grief session record
- Echoes archive entry
- grief overlay evidence packet when appropriate
- soft next-step suggestions

### Write-back law

- Default write-back is WB1 archive note in Echoes plus optional WB2 grief evidence packet.
- Last Light may inform grief overlays and intervention routing but may not directly rewrite archetype, shadow, or match score.
- Because grief can distort temporal readiness signals, downstream contracts should treat grief-linked outputs with contextual caps.

### AI permissions and constraints

- AI may write a healing-style reflection only with careful disclaimers and soft language.
- AI may not claim to channel the deceased, guarantee messages from beyond, or fabricate certainty about the afterlife.
- AI output must remain optional and clearly labeled as generated reflection.

### Safety, suppression, and fallback

- High-distress or crisis language should trigger SAFETY_HOLD and safer support routing.
- Media storage must honor deletion and privacy controls.
- Do not auto-surface Last Light content in normal matchmaking surfaces.

## 15. SAN_EXS - ExsAnonymous (Sanctuary Family)

ExsAnonymous is a Sanctuary healing and pattern-extraction family for processing prior relationships. It captures relational residue, repeated wounds, unfinished narratives, behavioral loops, learned defenses, attachment residue, and closure themes. It belongs under Sanctuary governance because it is reflective, emotionally loaded, and capable of seeding future interventions and clusters.

### Entry conditions

- User voluntarily opens ExsAnonymous or is gently routed after a breakup-related reflection, drift event, or recurring ex-linked pattern.
- ExsAnonymous may operate in staged depth levels so the user is not forced into a full processing session in one pass.

- Because prior-relationship content can be intense, safety and pacing rules are stricter than standard Mirror flows.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| EXS_INIT | Session created | Choose entry depth and target ex context | EX_REL_SELECT, QUICK_PASS, CANCELLED |
| EX_REL_SELECT | Relationship target chosen | Capture ex label and broad timeframe | STORY_CAPTURE |
| QUICK_PASS | Low-depth pulse check | Collect one-screen residue signal | PATTERN_REVIEW, COMPLETED |
| STORY_CAPTURE | Narrative input stage | Collect relational story fragments and key moments | PATTERN_EXTRACTION, PAUSED |
| PATTERN_EXTRACTION | System organizes signals | Map themes, wounds, defense patterns, lessons | AWAITING_AI, PATTERN_REVIEW, SAFETY_HOLD |
| PATTERN_REVIEW | User reviews extracted themes | Confirm, soften, reject, or save | CLOSURE_STEP, ARCHIVE_EXS, ACTIVE |
| CLOSURE_STEP | Optional release, reframe, or vow step | Capture closure act or future boundary | ARCHIVE_EXS, COMPLETED |
| ARCHIVE_EXS | Save ExsAnonymous packet | Persist ex-session artifacts and evidence refs | COMPLETED |

### Input payload contract

- ex_label or anonymous identifier
- relationship timeframe coarse fields
- optional relationship_type and significance level
- narrative fragments
- structured tags: wound_themes, lessons, repeated_conflicts, unfinished_threads
- optional freeform what still lingers field
- optional choice to keep private, archive, or allow AI context

### Primary outputs

- Ex session packet
- residue vector and healing themes
- cluster seeds such as overgeneralization, repetition, fear carryover, unfinished attachment residue, readiness overstatement, pace protection, and others where supported
- grief-linked or anger-linked companion evidence where present
- archive entry and optional future guidance prompts

### Write-back law

- ExsAnonymous writes WB1 archive notes by default and may write WB2 evidence packets when cluster or drift rule thresholds are satisfied.
- It may write WB3 AI context when the user consents to future coaching use.
- It may not directly lower a match score because the user had a painful history; only governed current-behavior evidence and approved overlays may travel downstream.
- ExsAnonymous is one of the richest sources for compassionate cluster interpretation, but its influence must remain capped and evidence-linked.

### AI permissions and constraints

- AI may help structure the story, summarize themes, and propose possible patterns the user may want to reflect on.
- AI must avoid deterministic statements like you always choose this kind of person unless the evidence path is explicit and softened.
- AI may not weaponize the user's ex-history in future match surfaces.
- Any generated theme must be rejectable by the user and stored with provenance.

### Safety, suppression, and fallback

- Abuse, trauma, coercion, or severe distress disclosures require careful escalation paths and content handling.
- Users must be able to keep an ExsAnonymous session fully private.
- Do not force closure rituals; some sessions are allowed to end as observation only.

### Implementation notes

- Recommended submode families: story capture, lesson extraction, residue scan, closure letter, pattern mirror, future-boundary vow.
- ExsAnonymous outputs are especially useful for Sanctuary-DM Reflection and future AI coaching, but only under consent.

## 16. SAN_DMREF - Sanctuary-DM Reflection

Sanctuary-DM Reflection is the post-conversation emotional processing module tied to a specific messaging thread or match dynamic. It exists to help the user notice how interactions affect them, not to spy on messages as a hidden surveillance engine.

### Entry conditions

- Triggered manually or after qualifying thread counts, conflict events, silence periods, or notable sentiment shifts if the user has consented to chat-linked reflection.
- Requires explicit message-analysis consent for any use of conversation-derived context.
- Should only ingest the minimum slice of conversation needed for reflection.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| DMR_INIT | Session starts with thread context | Load minimal metadata and optional snippets | FEELING_CHECK |
| FEELING_CHECK | User asked how interaction felt | Collect emotional response | PATTERN_LINK |
| PATTERN_LINK | System compares reaction to known patterns | Draft reflection and possible cluster links | AWAITING_AI, REVIEW_REFLECTION |
| REVIEW_REFLECTION | User reviews reflection | Save, dismiss, or route to Whisper or Mirror | COMPLETED, ACTIVE |

### Input payload contract

- thread_ref or match_context_id
- minimal message context if permitted
- user feeling rating and tags
- optional text reflection
- optional compare-to-prior state flag

### Primary outputs

- conversation reflection packet
- relationship dynamic hints
- possible cluster confirmations
- archive note and coaching route

### Write-back law

- DM Reflection may create WB2 evidence if message-linked thresholds are met and consent exists.
- It may update drift confidence or pair forecast context through governed downstream packets, never by bypassing the forecast contract.
- It should not store full raw chat bodies in Sanctuary logs when references suffice.

### AI permissions and constraints

- AI may summarize interaction patterns and suggest healthier next moves.
- AI may not provide manipulative strategy or hidden profile reading of the other person.
- AI must not infer private traits of the other user beyond what policy allows.

### Safety, suppression, and fallback

- If message-analysis consent is absent, downshift to self-report only mode.
- Use strict minimization on copied chat snippets.

## 17. SAN_TRUTH - Truth Signal

Truth Signal is the short daily or periodic insight layer. It takes governed evidence and produces a concise line intended to orient, not overwhelm. It is closer to a compass tap than a full reflection session.

### Entry conditions

- Shown only when enough signal exists or when a safe generic fallback line is available.
- May be generated after a session or as a daily pulse on Sanctuary home.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| TRUTH_READY | Signal can be generated | Select evidence bundle and copy template | TRUTH_RENDER |
| TRUTH_RENDER | Line generated | Display line and next-step CTA | COMPLETED |
| TRUTH_SUPPRESS | Not enough clean evidence or policy block | Show neutral fallback or nothing | AVAILABLE |

### Input payload contract

- evidence bundle ref
- copy template family

- user preference for tone if supported

### Primary outputs

- single insight line
- optional CTA into Mirror, Stillness, or ExsAnonymous

### Write-back law

- Truth Signal usually writes WB0 or WB1 only.
- It should not create new heavy evidence on its own; it renders existing governed context.

### AI permissions and constraints

- AI may generate phrasing, but copy must come from a controlled library and safety review.
- Truth Signal must avoid prophecy language, clinical claims, or partner shaming.

### Safety, suppression, and fallback

- If evidence is weak, suppress rather than fake depth.
- Keep tone grounded and non-absolute.

## 18. SAN_MM_GATE - Mirror Matches Reflection Gate

Mirror Matches Reflection Gate is a lightweight intention and readiness checkpoint before certain match surfaces are opened or after a match is presented. It is not a score calculator; it is a consentful reflective gate.

### Entry conditions

- Triggered by product policy before selected match interactions or after certain gated actions.
- Must be quick, clear, and not punitive.

### Local state machine

| State | When entered | Primary actions | Legal exits |
|---|---|---|---|
| MMG_INIT | Gate opened | Explain why a brief reflection is requested | INTENTION_CHECK |
| INTENTION_CHECK | Prompt shown | Collect quick intention or readiness response | DECISION |
| DECISION | Policy resolves | Open match surface, route to Sanctuary, or defer | COMPLETED |

### Input payload contract

- intention tags

- quick readiness self-check
- optional skip reason

### Primary outputs

- gate completion record
- route decision
- optional gentle module recommendation

### Write-back law

- May write WB4 gate state and WB1 small archive note if the user opts to keep it.
- Only repeated gate failures under explicit policy may create WB2 downstream context.

### AI permissions and constraints

- AI involvement should be minimal; templates usually suffice.
- Any generated language must remain short and non-judgmental.

### Safety, suppression, and fallback

- Do not turn a gate into emotional homework every time the user wants to act.

## 19. Module Interlock Rules Across Sanctuary

- Stillness may route into Mirror, Whisper, or Compass after regulation; it should not route directly into deep ExsAnonymous unless the user chooses it.
- Mirror may route into ExsAnonymous when the active wound clearly points backward, into Whisper when the urge is expression, into Ritual for decompression, or into DM Reflection for live relational context.
- ExsAnonymous may route back into Stillness or Ritual after heavy processing and may create future guidance hooks for Truth Signal.
- DM Reflection may call Whisper if the user wants to express something, or Mirror if the user needs to understand what the conversation stirred up.
- Ghost Mode affects entry surfaces and notifications for some modules but never blocks private Sanctuary reflection tools.
- Archive is retrieval and storage infrastructure; it should not become an active decision-maker.

## 20. AI Permission Matrix

| Module | Summarize | Ask follow-up | Infer patterns | Store context | Hard limit |
|---|---|---|---|---|---|
| Stillness | yes | light | no | limited | Never over- |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | interpret one regulation event. |
| Mirror | yes | yes | soft only | yes with consent | Hypotheses, not verdicts. |
| Whisper | yes | yes | light | yes with consent | No impersonation or manipulation advice. |
| Archive | yes | n/a | trend only | retrieval only | Do not replace raw source entries. |
| Compass | yes | light | trend only | yes | Keep quadrant labels controlled. |
| Ritual | light | light | no | limited | No unsafe instructions. |
| Ghost Mode | explain only | no | no | state only | No coercion. |
| Last Light | yes | light | very soft | limited | No supernatural claims. |
| ExsAnonymous | yes | yes | soft only | yes with consent | No weaponizing ex-history. |
| DM Reflection | yes | yes | soft only | yes with consent | No covert profiling of other user. |
| Truth Signal | generated line | no | render only | no | No prophecy language. |
| MM Gate | template-first | minimal | no | minimal | Keep it brief. |

## 21. Readiness, Forecast, Drift, and Cluster Interaction Rules

- Sanctuary may influence readiness and forecast only through governed downstream packets, never by direct score rewrites.

- Sanctuary evidence is especially useful for confidence, sufficiency, explanation, intervention routing, and suppression logic.
- Drift confirmations may be seeded by Compass, DM Reflection, Mirror, and ExsAnonymous, but formal drift transitions must honor the DriftEligibility appendix and rollup thresholds.
- Cluster seeds from Sanctuary remain low-confidence until supported by rule-pack evidence windows and cross-source confirmation where required.
- Grief-linked outputs from Last Light and ExsAnonymous should be context-aware so temporary mourning does not masquerade as stable incompatibility.
- No Sanctuary module may permanently alter archetype or shadow labels from a single session.

## 22. Safety and Moderation Requirements

- All Sanctuary modules must support SAFETY_HOLD with reviewed copy and escalation options when severe distress, self-harm, threats, coercion, or acute abuse disclosures arise.
- Modules with relationship-specific content must respect privacy minimization and never expose one user's private Sanctuary material to another user by default.
- Generated outputs must avoid diagnosis, prophecy, supernatural certainty, and manipulative strategy.
- Trust and safety review queues must receive structured references, not giant unstructured blobs whenever possible.
- ExsAnonymous, Last Light, and DM Reflection require especially strong retention and access controls because they can contain highly sensitive personal narrative.

## 23. Engineering Packet Templates

Every submode or future Sanctuary addition must ship with the following packet templates before production release:

- Module definition packet: module_id, version, owner_service, primary scope, legal entry surfaces, required permissions.
- State machine packet: states, transitions, timeouts, retry behavior, fallback behavior, completion criteria.
- Input payload schema: required and optional fields, accepted enums, max lengths, media rules, reference object rules.
- Output payload schema: saved objects, archive refs, evidence refs, route decisions, visibility scope.
- Logging packet samples: root session, event stream, error packet, moderation packet.
- Write-back matrix: WB class, destinations, thresholds, retention, deletion behavior, downstream consumers.

- AI behavior note: prompt family, allowed transforms, forbidden acts, provenance requirements.
- QA pack: happy path, sparse-input path, interrupted path, AI-failure path, safety-hold path, privacy-denied path.

## 24. Implementation Sequence

- Phase 1: implement shared Sanctuary shell, universal session object, event logging, and Archive plumbing.
- Phase 2: ship Stillness, Emotional Mirror, Whisper Window, and Ghost Mode because they provide immediate user value with manageable complexity.
- Phase 3: add Alignment Compass and Micro Rituals tied to the same event model.
- Phase 4: add ExsAnonymous and Sanctuary-DM Reflection after safety and consent layers are proven.
- Phase 5: add Last Light and Truth Signal once copy libraries and moderation controls are fully hardened.
- Phase 6: add Mirror Matches reflection gate and any advanced routing automation.

## 25. Final Contract Stance

Sanctuary is not a vague wellness corner. Under this specification it becomes a governed module family with shared runtime law, explicit packets, auditable transitions, and strict limits on what reflection may do to the score engine. ExsAnonymous is formally inside Sanctuary, which prevents duplicated logic, fragmented safety behavior, and conflicting write-back paths. As the broader Genesis OS stack matures, each submode should maintain its own packet appendix and QA examples beside this core contract.